

Semantic Communication Enabled 6G-NTN Framework: A Novel Denoising and Gateway Hop Integration Mechanism

Loc X. Nguyen¹, Sheikh Salman Hassan², *Member, IEEE*, Yan Kyaw Tun³, *Member, IEEE*, Kitae Kim⁴, *Member, IEEE*, Zhu Han⁵, *Fellow, IEEE*, and Choong Seon Hong⁶, *Fellow, IEEE*

Abstract—The sixth-generation (6G) non-terrestrial networks (NTNs) are crucial for real-time monitoring in critical applications like disaster relief. However, limited bandwidth, latency, rain attenuation, long propagation delays, and co-channel interference pose challenges to efficient satellite communication. Therefore, semantic communication (SC) has emerged as a promising solution to improve transmission efficiency and address these issues. In this paper, we explore the potential of SC as a bandwidth-efficient, latency-minimizing strategy specifically suited to 6G satellite communications. The existing SC methods have demonstrated efficacy in direct satellite-terrestrial transmissions; however, they still encounter certain limitations. Specifically, some ground users (GUs) experience poor signal-to-noise ratios (SNR), making direct satellite communication challenging. To address these issues, we propose a novel framework that optimizes gateway hop-relay selection for GUs with low SNR and integrates gateway-based denoising mechanisms to ensure high-quality-of-service (QoS) in satellite-based SC networks. This approach directly mitigates distortion, leading to significant improvements in satellite service performance by delivering customized services tailored to the unique signal conditions of each GU. Our findings represent a critical advancement in reliable and efficient data transmission from the Earth observation satellites, thereby enabling fast and effective

responses to urgent events. Simulation results demonstrate that our proposed strategy significantly enhances overall network performance, outperforming conventional methods by offering tailored communication services based on specific GU conditions.

Index Terms—6G, NTN, semantic communication, hop-based relaying, satellite access networks, denoising, GU fairness.

I. INTRODUCTION

THE non-terrestrial networks (NTNs), specifically satellite access networks (SANs), have emerged as a vital solution for achieving large-scale connectivity and seamless coverage, which are critical for next-generation wireless networks [1]. Unlike terrestrial communication systems, NTNs face challenges such as high launch costs, dependence on line-of-sight (LoS) transmission, and long propagation distances. To address these challenges, low Earth orbit (LEO) satellite¹ systems like Iridium, Globalstar, OneWeb, Starlink, and Telesat have been deployed [2]. These systems initially used transparent satellites that amplified signals but now rely increasingly on regenerative satellites [3]. Regenerative satellites function as base stations with onboard processing capabilities, such as modulation and coding, and support inter-satellite links within LEO constellations—an essential feature for global networking. While the Autonomous Aerial Vehicle has energy-constrained and limited computation capabilities and thus requires a well-designed solution to provide high-quality services for users [4], [5], the satellites offer extensive coverage and powerful processing capabilities. Nevertheless, satellites are not without challenges.

The high mobility of satellites leads to performance degradation, prompting research into improving the reliability and adaptability of NTNs [6], [7]. Techniques like robust beamforming, adaptive modulation, and interference mitigation are being explored to enhance transmission performance. Large satellite constellations, combined with the diversity of services and wide coverage areas, introduce further complexity. Additionally, aggressive frequency reuse in satellite systems, aimed at increasing spectrum efficiency, exacerbates co-channel interference (CCI). Addressing the unpredictable nature of CCI, caused by satellite movement, remains an ongoing challenge. While hybrid automatic repeat request (HARQ) mechanisms are sometimes employed to manage transmission errors, these

¹Hereinafter, LEO satellite will be considered as satellite unless otherwise stated.

Received 24 September 2024; revised 29 December 2024 and 15 March 2025; accepted 4 June 2025. Date of publication 17 June 2025; date of current version 12 December 2025. This work was partially supported by Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korea Government (MSIT) (No. 2019-0-01287, Evolvable Deep Learning Model Generation Platform for Edge Computing), IITP grant funded by Korean MSIT (Artificial Intelligence Innovation Hub) under Grant RS-2021-II212068, MSIT (No. RS-2022-00155911, Artificial Intelligence Convergence Innovation Human Resources Development (Kyung Hee University)), MSIT under the Convergence security core talent training business support program (IITP-2023(2023)-RS-2023-00266615) supervised by the IITP, the Information Technology Research Center (ITRC) grant funded by MIST (IITP-2025-RS-2023-00258649, 50%), NRF grant funded by the Korea Government (MSIT) (No. RS-2023-00207816), NRF grant funded by the Korea Government (MSIT) (No. RS-2024-00352423) and IITP grant funded by the Korea Government (MSIT) (No. RS-2024-00509257, Global AI Frontier Lab). The associate editor coordinating the review of this article and approving it for publication was A. Zaidi. (*Corresponding authors: Kitae Kim; Choong Seon Hong.*)

Loc X. Nguyen, Sheikh Salman Hassan, Kitae Kim, and Choong Seon Hong are with the Department of Computer Science and Engineering, Kyung Hee University, Yongin-si, Gyeonggi-do 17104, Republic of Korea (e-mail: xuanloc088@khu.ac.kr; salman0335@khu.ac.kr; glideslope@khu.ac.kr; cshong@khu.ac.kr).

Yan Kyaw Tun is with the Department of Electronic Systems, Aalborg University, 2450 Aalborg, Denmark (e-mail: ykt@es.aau.dk).

Zhu Han is with the Department of Electrical and Computer Engineering, University of Houston, Houston, TX 77004 USA, and also with the Department of Computer Science and Engineering, Kyung Hee University, Seoul 446-701, South Korea (e-mail: hanzhu22@gmail.com).

Digital Object Identifier 10.1109/TWC.2025.3578010

are limited by the substantial and variable delays between satellites and ground stations [8]. As a result, deep learning has been proposed as a solution to tackle these complex challenges in NTN [9]. Despite the support of DL solutions for resource and network management problems, traditional communication still has difficulty transmitting large amounts of data from satellite systems within a limited timeframe [10].

Semantic communication (SC) has shown significant potential in reducing bandwidth consumption and improving performance in challenging channel conditions [11]. By focusing on transmitting the meaning (semantics) of data rather than raw data, SC aligns with the vision of *Intelligence as a Service* in 6G networks. Recent advances, such as high computation capabilities and hardware improvement [12], have demonstrated the utility of SC in addressing NTN-specific challenges, such as satellite computation offloading, wireless UAV control, and remote spectrum sensing. However, further research is required to enhance the integration of SC into NTNs [13]. The SC prioritizes task-oriented performance over traditional bit-level accuracy and has proven effective in addressing connectivity and low-latency requirements of 6G. This approach is made possible by advancements in deep learning models, which enable efficient embedding of semantic information into the communication process [14]. The integration of SC into NTNs presents an innovative path forward, though significant challenges remain in optimizing satellite transmission using this paradigm.

This paper explores the potential of integrating SC with SANs, a powerful approach that focuses on conveying the meaning (semantics) of information rather than just the data itself [15]. This integration offers several advantages, i.e., SC has the potential to reduce communication overhead. Additionally, addressing the disparities in communication channel quality of ground users (GUs) through integration with terrestrial networks can further guarantee high-quality service for all communication participants [16]. These advancements pave the way for a global *Intelligence as a Service* network, fostering a more connected and intelligent world. The concept proposed in [17], where collaboration between distributed nodes combines information from multiple perspectives and leverages network processing capabilities for inference tasks, is inspiring. Specifically, in our paper, a terrestrial gateway can be leveraged by its massive computing and unlimited power to assist the user under high noise amplitude with a hopping connection. However, traditional hop-relaying techniques in satellite communication face limitations like high bandwidth consumption and error accumulation over long distances. Therefore, we propose a novel hopping connection for NTNs based on SC² through the ground gateway, where its role includes receiving the signal, decoding, denoising, and finally transmitting the signal to the destination user.

To the best of our knowledge, we are the first to investigate the application of SC in satellite-based hop relay networks. Our approach considers both direct access for GUs with favorable signal conditions and selective hop relaying for enhanced reliability and fidelity. While most existing SC research

focuses on single-hop terrestrial communication, this paper ventures into a novel concept where hop-relaying satellite networks with denoising are implemented. We acknowledge valuable contributions to SC in terrestrial networks, such as physical-layer network coding with semantic decoders [18], cooperative transmission with half-duplex relays [19], and autoencoder-based SC systems with semantic forward protocols [20]. This research seeks to bridge the gap by investigating SC in satellite hop relay networks. This necessitates satellite-based relaying to ensure remote coverage and avoid deep fading through line-of-sight (LoS) connections. Furthermore, we investigate SC on hop relay networks and evaluate the impact of hop forwarding on the fidelity of semantic information. The major contributions of this work are summarized as follows:

- In this paper, we consider an SC system tailored for satellite architectures that serve multiple GUs. Recognizing that the high altitude of satellites can impact the SNR for GUs, we propose a gateway-assisted approach to enhance communication quality for these GUs.
- We leverage the high-computing capacity and extensive resources in received antenna power of the gateway in NTN communications to mitigate environmental noise and meet the high QoS demands of GUs. Specifically, the received power at the receiver influences the signal-to-noise ratio (SNR), with the gateway delivering a higher quality signal compared to mobile GU devices.
- Next, we formulate an optimization problem to minimize the total communication overhead for all GUs while ensuring their reconstruction quality requirements are met. The objective of the formulated problem is accomplished by optimizing the pathway for GUs, i.e., whether direct or hopping communication and determining the optimal allocation of carrier frequencies. To address this problem efficiently, we introduce a two-stage discrete whale optimization algorithm (D-WOA), which delivers a sub-optimal solution within a polynomial time frame, crucial for adapting to the satellite's dynamic environment.
- Finally, we evaluate the system's performance with a standard satellite dataset, comparing it against various optimization benchmarks. Additionally, we assess the effectiveness of the gateway-assisted hop approach in producing high-quality images in a high-noise environment.

The remainder of this paper is organized as follows. Section II summarizes the related work, while Section III provides the system model. Section IV details the problem formulation, and Section V introduces the proposed solution, including the network structure and training process. Section VI showcases simulations performed on the satellite-based dataset. Finally, Section VII concludes the paper.

II. RELATED WORKS

A. Non-Terrestrial Networks

The terrestrial communication infrastructure undeniably offers advanced services for ground users (GUs); however, its deployment is costly. As a result, such infrastructure is

²In this research article, NTNs refer specifically to SANs.

typically concentrated in urban areas, with limited availability in rural regions [21]. To achieve seamless 6G connectivity for both areas, NTN is gaining attention for their flexible, high-throughput, and cost-effective deployment, especially in remote or harsh environments. Specifically, the development of satellite constellations and UAV base stations (BSs) promises high-quality network connections and diverse services for remote GUs [22], [23], [24], [25]. In [22], the authors considered a group of UAVs to guarantee efficient data sharing among vehicles by controlling the UAV movements. Similarly, a more extensive coverage network is considered in [23], where the proposed Space-Air-Ground-Integrated Networks (SAGINs) offers massive computing to enhance the quality of different vehicle services in remote areas. Inspired by the edge computing enhancement in SAGINs, the study in [24] introduces a novel approach that utilizes SANs along with LEO and cube satellites as multi-access edge computing (MEC) servers for data offloading and computation services. It focuses on optimizing offloading task selection and resource allocation to tackle the challenges of joint delay and rental price minimization. Meanwhile, [25] addressed the challenge of managing capacity across three satellite layers (geosynchronous, medium, and low earth orbit) by proposing a Q-learning framework along with a low-complexity algorithm.

B. Semantic Communication

Tremendous attention has been shifted from traditional wireless communication systems to semantic communication, which can effectively transmit the message's meaning only [26]. The growth of SC has been accelerated by advances in deep learning, such as Convolution Neural Networks (CNNs) and Vision Transformers (ViT) [27], which can extract the semantic features within the data, transmit and recover successfully. SC covers a range of areas, like goal-oriented communication, AI-generated content (AIGC), and data reconstruction. In this paper, we focus on data reconstruction, especially for image transmission, and provide a detailed review of related work in this direction [28], [29], [30], [31], [32]. By jointly optimizing the source encoding with the channel encoding process, the work [28] has demonstrated its success in embedding the semantic meaning into the transmission and also eliminating the *cliff effect* of traditional communication. Building on this work, the authors leveraged the channel feedback information by incorporating it into the transmission process and successfully improved the reconstruction quality [29]. The authors in [30] were the first to apply the attention mechanism proposed in [33] to the field of semantic image communication, effectively quantifying important features while filtering out redundancies. As in [32], a novel semantic framework is proposed to predict the quality of the received image at the receiver site, thus adjusting the compression ratio by selecting the first R-proportion semantic features to transmit. While the idea of PADC in [32] is interesting and effective in improving bandwidth efficiency, it requires zero-padding operation at the receiver. This process results in a significantly larger padded feature map, which must then be decoded by the receiver's decoder, thereby increasing computational complexity and energy consumption.

In contrast, our proposal eliminates the need for additional processing at the GUs by utilizing the hopping connection through the ground gateway. Furthermore, we adopted Swin-Transformer [34] as our semantic encoder/decoder due to its impressive performance and computation efficiency.

C. Semantic Communication Based Non-Terrestrial Networks

In this sub-section, we provide an overview of the literature related to semantic communication in Non-Terrestrial Networks [35], [36], [37], [38]. In [35], the authors proposed a scenario where a UAV sensor is employed to collect sensing data from an unobservable moving primary user (PU) for the spectrum sensing application. Recognizing the limited computing capability and high latency, the authors adopted the semantic communication from the UAVs to the BS, which only transmits features of the detection task. Then, a deep reinforcement learning (DRL) solution is proposed to help the UAV track the moving PU. A broader coverage scenario is considered in [36] to help resource-limited GUs by deploying the LEO satellites equipped with a satellite-borne edge cloud. In the paper, the authors investigated how the deployment of SC led to the trade-off between the communication benefit and computation resource for the coder updating requirements. Therefore, they divided the system into two branches: in-maintenance (updating the parameters of the coder in a federated learning manner) and in-service (executing the computation task), which guarantees the feasibility of computing service and highly efficient communication. In the work [37], a UAV is leveraged as a relay to assist the data transmission from the satellite to the ground terminal (GT) and relay through the UAV. In addition, they addressed the energy efficiency of the proposed system by probabilistic semantic communication (PSC), where the satellite transmits a portion of the message, and the GT fills in the missing information by leveraging a shared knowledge graph.

Unlike previous works where the satellite serves as the source of data, the satellite in [38] acts as a relay node, connecting the user terminal and the gateway to extend service coverage and enhance transmission efficiency in challenging environments. The authors proposed a foundation model (FM) to segment an image into C categories features, which reduces the signal length and then input it into a semantic encoder before transmitting to the gateway through the satellite links. At the gateway, a combination of the semantic decoder and FM-based generator is proposed to generate an image that resembles the semantic meaning of the original image. In our recent study [39], we present a novel SC-based approach that successfully adjusts transmission length and optimizes latency in the satellite imagery downlink for Earth observation, focusing solely on direct communication scenarios.

D. Challenges in the Literature and Rationale for Proposal

The studies in II-A have underscored the increasing significance of NTN in mobile communication, particularly highlighting their extensive coverage and seamless connectivity [15], [40]. In II-B and II-C, we have provided comprehensive work related to the potential of semantic

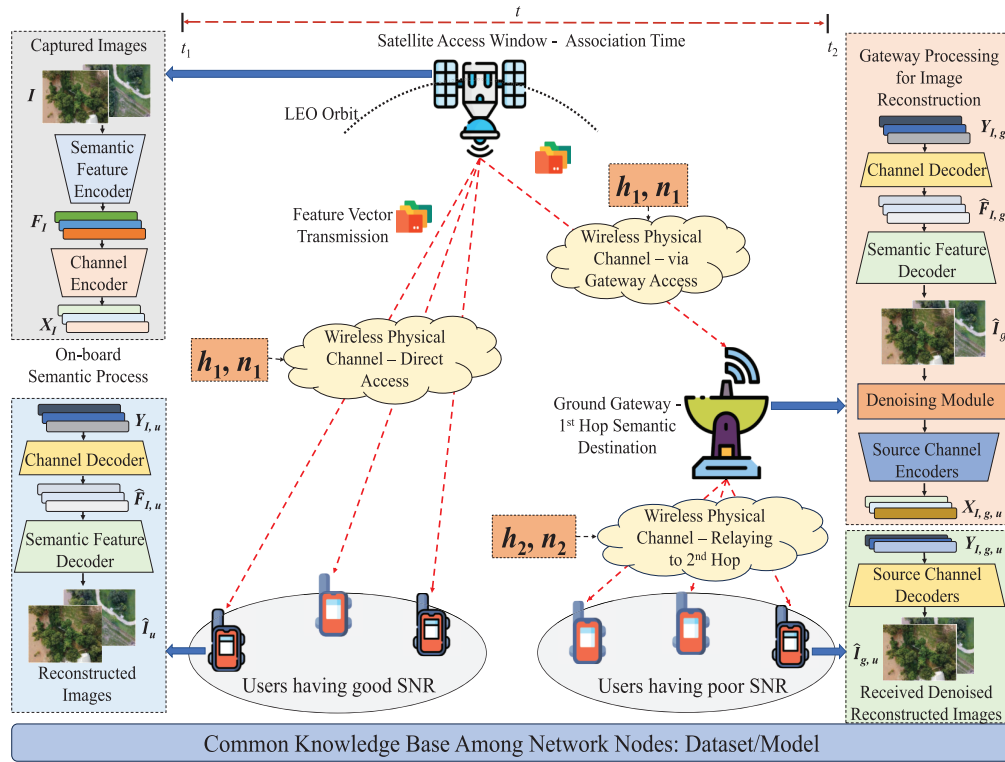


Fig. 1. The architectural framework for facilitating satellite-based semantic communication to the diverse remote GUs.

communication and the benefits of implementing SC-based NTN architecture sequentially. However, they failed to address several critical challenges, including the low-computing complexity of GUs, the dynamic requirements for the QoS among GUs, and finally, the difference in channel conditions of GUs. In response, our proposed work explores the application of the state-of-the-art transformer technique, Swin Transformer, to efficiently decode the received semantic features while maintaining low computational complexity. Additionally, we explore intelligent resource management strategies tailored to individual GU conditions, guaranteeing the QoS and efficient SC in NTN.

III. SYSTEM MODEL

A. Network Model

As illustrated in Fig. 1, the proposed SC system comprises satellite s , gateway g , and a set $\mathcal{U}$ of U GUs, i.e., $u, u' \in \mathcal{U}$, where u -indexed GUs are those whose signal-to-noise ratio (SNR) favorable for direct satellite-based SC and u' -indexed GUs are those which cannot directly associate with satellite for SC due to their poor SNR, thus $u \cup u' = \mathcal{U}$. Let's define another joint set $\mathcal{M}$ of M composed of all good SNR GUs and a gateway known as direct ground nodes. Here, we consider the captured image reconstruction task as the service of the system, where the satellite encodes the semantic features within the image and transmits them to GUs within the network domain. Then, the GUs try to reconstruct the original image from the received signal, and each GU will have different requirements for the QoS, such as time or image quality. Due to the mobility of the satellite and

the dispersed locations of the GUs, the connections between the satellite and GUs are dynamic, impacting the channel quality. When the channel conditions between the satellite and a GU are favorable, the GU can achieve high-quality image reconstruction. Conversely, at elevated noise levels, the decoders at the GUs struggle to reconstruct images with high fidelity. To overcome this challenge, we propose utilizing a ground gateway as an intermediary hop and denoising node in the SC process. Specifically, the satellite transmits the image to the gateway, which benefits from virtually unlimited computational power and energy resources. At the gateway, a DL-based denoising model is employed to mitigate the noise effects before relaying the refined semantic signal to the intended GUs. Additionally, when the gateway itself serves as the target GU, it can fully reconstruct the image with high accuracy.³

B. Satellites-Based Semantic Feature Extraction Model

Similar to standard SC systems, the transmitter at the satellite comprises a semantic encoder and a channel encoder, which will be jointly optimized. The semantic encoder is responsible for extracting the important features within the image I , which significantly reduces the transmitted data and saves more bandwidth resources. We denote the size of the original image as n and $I \in \mathbb{R}^n$, the value of n is calculated by the multiplication among its height, width, and the number

³It is assumed that since the gateway possesses substantial computing power, it is more efficient to reconstruct the image/data at the gateway rather than relying on low SNR GUs. This approach ensures optimal utilization of network resources and enhances service quality.

of color channels. The semantic features of the image I can be denoted as:

$$F_I = E_\alpha(I), \forall I \in \mathbb{R}, \quad (1)$$

where $E_\alpha(\cdot)$ denotes the semantic encoder and α is its parameter. The semantic features are then compressed by the channel encoder, which lowers the transmission length and protects them from the physical noise from the environment. To effectively combat the noise, the channel encoder will take the Γ SNR value as a condition in the process and output the encoded semantic symbol as:

$$X_I = C_\beta(F_I|\Gamma), \forall \Gamma \in \text{SNR}, \quad (2)$$

where $C_\beta(\cdot)$ indicate the channel encoder and its corresponding network parameters β . The size of the encoded semantic symbol X_I is expressed as b and the ratio between the size of the encoded semantic symbol and the original image size is calculated by b/n .

C. Satellites-Based Semantic Feature Transmission Model

We consider an orthogonal frequency-division multiple access (OFDMA) due to its flexibility in allocating limited network resources. This allows satellites to be assigned subcarriers based on their specific data requirements, efficiently handling dynamic GU channel conditions. Additionally, OFDMA inherently avoids inter-GU interference within the spectrum. A set $\mathcal{K}$ of K subcarriers is considered to allocate each GU if its SNR is above the QoS threshold (i.e., good SNR). Otherwise, the gateway will receive this semantic signal and will process and relay it to the end GUs after the reconstruction and denoising process. Let $\gamma_{s,m}^k$ denote the subcarrier assignment variable for each GU u or gateway g is:

$$\gamma_{s,m}^k \in \{0, 1\}, \quad (3)$$

i.e.,

$$\gamma_{s,m}^k = \begin{cases} 1, & \text{if subcarrier } k \text{ utilized by any direct} \\ & \text{connection, i.e., GU } u \text{ or gateway } g, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

We also provide the condition that only one subcarrier k is assigned to GU u or gateway g at each time slot to avoid network interference:

$$\sum_{m=1}^M \gamma_{s,m}^k \leq 1. \quad (5)$$

Similarly, each node m can utilize at most a single subcarrier, which is given as follows:

$$\sum_{k=1}^K \gamma_{s,m}^k \leq 1. \quad (6)$$

Thus, the downlink rate from the satellite to the GU u or gateway g on subcarrier k to transmit semantic feature information $F_{I,k}$ as:

$$R_{s,m}^k = B_{s,m}^k \log_2(1 + \Gamma_{s,m}^k), \quad \forall m \in \mathcal{M}, k \in \mathcal{K}, \quad (7)$$

where $B_{s,m}^k$ is the subcarrier bandwidth. Moreover, the SNR Γ between the satellite and the ground node m can be defined as follows:

$$\Gamma_{s,m}^k = G_s G_m P_s \left(\frac{c}{4\pi d_{s,m} f_{s,m}^k N_0} \right)^2, \quad (8)$$

where G_s and G_m denote the wireless power antenna gains for the satellite and GU/gateway, respectively. P_s is the satellite transmit power, c indicates the speed of light, $f_{s,m}^k$ denotes the carrier frequency, $d_{s,m}$ denotes the distance between the satellite s and the GU u or gateway g , and N_0 denotes the spectral density of the noise power to capture the interference generated by the satellites in other constellations which utilize the same bandwidth spectrum and subcarrier k .

D. GU Semantic Feature Reception and Reconstruction

The GUs can receive the semantic encoded feature from the satellite in two pathways: direct connection and indirect connection via the gateway. This pathway will be determined by environmental noise and the GU demand for image quality. In both cases, the GU decodes the encoded semantic symbol using its semantic and channel decoders.

1) *Direct GU Connections*: With the encoded semantic symbol transmitted through a wireless channel, environment noise is inevitable, and the received signal at the GU is:

$$Y_{I,u} = X_I + N_u \quad (9)$$

where N_u denotes the channel noise, whose element follows the Gaussian distribution $\mathcal{N}(0, \sigma^2 I)$. Here we consider both the additive white Gaussian noise (AWGN) channel and the Rayleigh fading channel.

$$Y_{I,u} = H_{I,u} X_I + N_u, \quad (10)$$

where $H_{I,u}$ denotes the fading coefficient of the Rayleigh channel between communication devices. Initially, GU u decodes the received signal with the channel decoder:

$$\hat{F}_{I,u} = C_{\phi_u}^{-1}(Y_{I,u}), \quad (11)$$

where $C_{\phi_u}^{-1}(\cdot)$ represent the channel decoder at GU u and the corresponding parameters ϕ_u . Then the semantic decoder takes the semantic feature of the image I as input and reconstructs the original image as:

$$\hat{I}_u = D_{\theta_u}(\hat{F}_{I,u}), \quad (12)$$

where $D_{\theta_u}(\cdot)$ denote the semantic decoder of GU u and its parameter θ_u .

2) *Indirect GU Connections via Gateway*: In extremely harsh environmental conditions, GUs cannot achieve high-quality image reconstruction due to their models and power limitations. To address this, the satellite actively transmits signals to a gateway, relaying these signals to the GU. As shown in the equation (8), with the same environment noise and approximate distance, the gateway is equipped with powerful antenna gains, leading to better-received signals than the GU. Additionally, leveraging the gateway's abundant resources, we propose a DL-based denoising model to mitigate noise before relaying the signal to the GU u' . Details of the denoising

model are provided in III-E. Here, we focus on the link between the gateway g and the GU u' . Lets $\chi_{g,u'}^j$ denotes the subcarrier assignment variable for each GU u' is:

$$\chi_{g,u'}^j \in \{0, 1\}, \quad (13)$$

i.e.,

$$\chi_{g,u'}^j = \begin{cases} 1, & \text{if subcarrier } j \text{ utilized by any } u \text{ indirect GU,} \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

We also provide the condition that only one subcarrier j is assigned to each GU at each time slot to avoid network interference:

$$\sum_{u'=1}^{U-u} \chi_{u'}^j \leq 1. \quad (15)$$

Similarly, each GU u' can utilize at most a single subcarrier, which is given as follows:

$$\sum_{j=1}^J \chi_{u'}^j \leq 1. \quad (16)$$

The downlink rate from the gateway g to the GU u' on subcarrier j to transmit the data of reconstructed image I as:

$$R_{g,u'}^j = B_g^{u'} \log_2(1 + \Gamma_{g,u'}^j), \quad \forall u' \in \mathcal{U} - u, j \in \mathcal{J}, \quad (17)$$

where $B_g^{u'}$ is the subcarrier bandwidth. Moreover, the SNR Γ between the gateway and the GU can be defined as follows:

$$\Gamma_{g,u'}^j = G_g G_{u'} P_g \left(\frac{c}{4\pi d_{g,u'} f_{g,u'} N_1} \right)^2, \quad (18)$$

where G_g and $G_{u'}$ denote the wireless power antenna gains for the gateway and GU, respectively. P_g is the gateway transmit power, c indicates the speed of light, f_c denotes the carrier frequency, $d_{g,u'}$ denotes the distance between the gateway and the GU.

E. Integrated Hop Gateway for Processing and Relaying

In contrast to the limitations on various aspects, such as energy and computing capacities of the GUs, the gateway has a powerful computing capability and unlimited energy supply. Therefore, the assistance provided by the gateway to the GUs is twofold: leveraging a powerful antenna to receive the signal and providing the denoising mechanism with its unlimited computing resources. Equation (8) shows that the SNR is influenced not only by environmental noise but also by the transmitter and receiver power and the distance between them. The GU is constrained by the limited power and hardware design for the mobile antenna, which eventually experiences low SNR in harsh physical environments and over long distances. On the other hand, when the gateway receives the signal under the same noise conditions (N_0) and a similar distance ($d_{s,u}$), the gateway's powerful antenna significantly enhances the SNR. The improvement in SNR indicates that the gateway can capture a signal more closely resembling the original transmitted signal than the GU device. Additionally, a better SNR between the satellites and the

gateway enables a faster transmission rate, which is essential for time-sensitive applications such as earth observation for disaster response: tsunamis, wildfires or tracking moving target [41]. Furthermore, utilizing its vast computing resources, the gateway can employ a deep learning-based denoising module to reduce environmental noise before relaying the signal to the GU, introducing extra computation costs from the gateway side. However, compared to the benefits it provides for the GUs and its unlimited resources, there will be more advantages than the number of drawbacks. Besides, the gateway can be a targeted receiver.

1) *Signal Relay Mode for the GUs*: Under this mode, the gateway first decodes the received signal by employing its joint source-channel decoder.

$$\hat{I}_g = D_{\theta_g}(C_{\phi_g}^{-1}(Y_{I,g}|\Gamma)), \quad (19)$$

where $Y_{I,g}$ denotes the received signal and $\hat{I}_g$ is the reconstructed image at the gateway. This image is reconstructed here, which can be beneficial not only to the high powers of the antenna here but also to the computing resource. To be specific, a denoising model refines the reconstructed image before putting it through the encoding process and transmit to the receiver. The denoising model requires a high computing capacity to implement, which is infeasible for mobile GUs. This process can be denoted as the following equation:

$$\hat{I}_{\text{den}} = \text{DEN}_{\eta}(\hat{I}_g), \quad (20)$$

where $\text{DEN}_{\eta}(\cdot)$ denote the denoising model with the parameter η . After finishing the denoising process, the gateway transmits the denoised signal to the GU, which also counters the noise from the environment. However, the signal will be less severely affected due to the relatively small distance compared to the satellite-GU distance. The encoding process at the gateway is represented as:

$$X_{I,g,u} = C_{\beta_g}(E_{\alpha_g}(\hat{I}_{\text{den}})|\Gamma), \quad (21)$$

where $X_{I,g,u}$ is the encoded signal at the gateway. GU u received the signal $Y_{I,g,u}$, which also experiences the environment noise and follows the formula as equation (9). Then, GU u decodes the signal to acquire the image.

2) *Semantic Image Reconstruction*: If the gateway is the intended recipient, it can easily reconstruct the image using the joint source-channel decoder. Additionally, it's worth noting that the denoising process is optional, which depends on the GU's and gateway's quality requirements.

IV. LATENCY MINIMIZATION PROBLEM FORMULATION

Based on the encoded semantic features obtained in (2), the achievable data rates calculated in (7), we can calculate the latency of directly transmitting the features from satellite s in subcarrier k to the GUs or gateway m is:

$$t_{s,m}^k = \frac{l_X}{R_{s,m}^k}, \quad \forall m \in \mathcal{M}, \quad (22)$$

where l_X denotes the length of the transmitted data. For GUs who receive the signal through the gateway indirect paths, the latency is the sum of two transmission processes: *latency from*

satellite to gateway, latency from gateway to GU. The latency is calculated as:

$$t_{s,u'} = \frac{l_X}{R_{s,m}^k} + \frac{l_{X_g}}{R_{g,u'}^j}, \forall u' \in \mathcal{U} - u, m = g, \quad (23)$$

where $R_{s,m}^k$ is only that satellite s is connected with gateway m via subcarrier k for relaying towards bad SNR GUs u' , l_{X_g} denotes the transmitted length from gateway to GUs u' .

Leveraging the aforementioned system model, we aim to minimize the average latency for transmitting semantic features from satellite s to GUs based on the optimization of expected received image quality $\mathbb{E}(\text{PSNR}_k(\mathbf{X}))$. However, this optimization must adhere to the network model's practical constraints. Specifically, sufficient semantic information must be transmitted within the access window to allow the direct access GUs or gateway to reconstruct the original send image. Additionally, reliability and SC-QoS must be ensured by guaranteeing a minimum PSNR threshold that meets the desired image recovery quality. Under these assumptions, the average latency from satellite to gateway can be given as:

$$\Psi_1(l_X, \gamma) = \frac{1}{M} \sum_{m \in \mathcal{M}} \sum_{k \in \mathcal{K}} \gamma_{s,m}^k t_{s,m}^k. \quad (24)$$

Similarly, the average latency from the gateway to GU can be given as:

$$\Psi_2(l_X, \chi) = \frac{1}{U-u} \sum_{u' \in \mathcal{U}-u} \sum_{j \in \mathcal{J}} \chi_{g,u'}^j t_{g,u'}^j. \quad (25)$$

Then the total latency O among all GUs can be calculated as:

$$O(S, G, U, \gamma, \chi) = \Psi_1 + \Psi_2. \quad (26)$$

where $S = (\alpha, \beta)$ comprise the parameters of the semantic and channel encoders at the transmitter, while $G = (\theta_g, \phi_g, \eta, \alpha_g, \beta_g)$ and $U = (\theta_u, \phi_u)$ denotes the network parameters of the gateway and GU u , respectively.

The latency minimization problem under the recovered image quality optimization and loss function minimization can be formulated as:

$$\mathbf{P1:} \min_{S, G, U, \gamma, \chi} O(S, G, U, \gamma, \chi) \quad (27a)$$

$$\text{s.t. } \gamma_m^k \in \{0, 1\}, \quad \forall k \in \mathcal{K}, \quad \forall m \in \mathcal{M} \quad (27b)$$

$$\sum_{m=1}^M \gamma_m^k \leq 1, \quad \forall k \in \mathcal{K} \quad (27c)$$

$$\sum_{k=1}^K \gamma_m^k \leq 1, \quad \forall m \in \mathcal{M} \quad (27d)$$

$$\chi_{u'}^j \in \{0, 1\}, \quad \forall j \in \mathcal{J}, \quad \forall u' \in \mathcal{U} - u \quad (27e)$$

$$\sum_{u'=1}^{U-u} \chi_{u'}^j \leq 1, \quad \forall j \in \mathcal{J} \quad (27f)$$

$$\sum_{j=1}^J \chi_{u'}^j \leq 1, \quad \forall u' \in \mathcal{U} - u \quad (27g)$$

$$\mathbb{E}(\text{PSNR}_u(\mathbf{X})) \geq \Psi_u, \quad \forall u \in \mathcal{U} - u' \quad (27h)$$

$$\mathbb{E}(\text{PSNR}_{u'}(\mathbf{X})) \geq \Psi_{u'}, \quad \forall u' \in \mathcal{U} - u \quad (27i)$$

$$0 \leq \gamma^k t_{s,m} \leq T_m, \quad \forall m \in \mathcal{M}, \quad (27j)$$

where constraints (27b), (27c), and (27d) guarantee that the satellite s will assign only a single subcarrier k to high-end ground nodes j (i.e., direct GU u or gateway g) for feature transmission. Every subcarrier could only be utilized by a single ground node m at each time slot for semantic features reception. Constraints (27e), (27f), and (27g) are second-stage constraints that guarantee that the gateway g will assign only a single subcarrier j to low-end ground nodes u' for denoised feature transmission, and every subcarrier could only be utilized by a single ground node u' at each time slot for denoised semantic features reception. Both constraints (27h) and (27i) ensure that the semantic features should be reliable and meet the SC-QoS threshold for every GU in a set $\mathcal{U}$. Constraint (27j) ensures the communication efficiency of satellites within the access window, i.e., the coverage period of the satellite s to ground nodes m . It can be observed that the formulated problem is MINLP, which is difficult to solve due to its being NP-hard. To address this problem, we provide a solution approach in the next section.

V. PROPOSED ALGORITHM

As shown in the formulation, the proposed problem depends on the deep learning parameters of the encoders, decoder, and denoise modules, which is similar to most other SC studies. Thus, optimizing the parameters of these modules is the top priority so that the SC system can effectively extract the image feature and efficiently transmit it over a wireless environment. Then, based on the optimized parameters of all the modules and the GU requirements, the transmit pathway will be decided to meet the demands of all GUs within the lowest communication time.

A. Parameters Optimization of Semantic Communication

In this subsection, we introduce the architecture of each encoder/decoder for the channel or source coding process, the denoising model, and finally, the optimization algorithm. To make it easier for the reader to follow, we will refer to the encoder/decoder pair as the coder from now on.

1) *Source Coder*: The semantic feature extraction from the raw data has been considered as a difficult task, necessitating the assistance of robust DL models. As a result, various architectures from simple to high complex have been proposed to extract the semantic features, such as conventional CNNs [28], Attention-assisted CNNs [30], [31], [32], and Transformer [42], [43], [44]. The transformer models have achieved impressive performance in source encoding/decoding compared to other architectures; however, they had drawbacks, the major one being the high-computing complexity. The high computing capacity is also not a feasible factor for the GUs, which motivates us to adopt the Swin Transformer as the backbone architecture for the semantic coder. The model can outperform the normal ViT in various vision tasks while demanding low computation resources [34], which particularly fit the GU device.

Swin Transformer first divides an image into non-overlapping local windows; each window contains $M \times M$ patches and applies the transformer mechanism on each window. The local self-attention within each image window costs

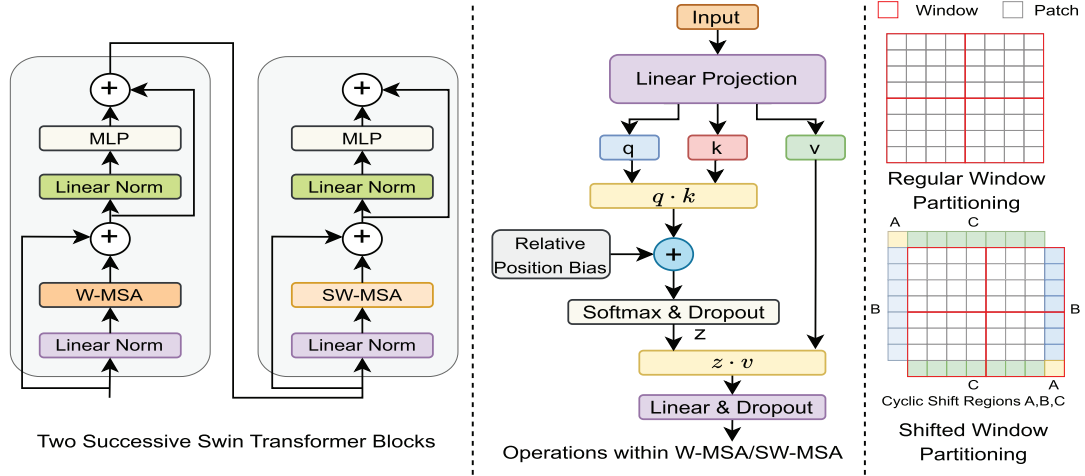


Fig. 2. In the figure, we illustrate the architecture between two Swin Transformer blocks, operations within the W-MSA/SW-MSA, and finally, the difference between regular and shifted window partitioning.

significantly lower computing operations than the global attention proposed in the Vision Transformer (apply the attention mechanism among all patches within the image). In addition, to create the connection among each window, every normal window is followed by a shifted window, which is implemented to capture the dependency correlation across windows. The shifted window partitioning maintains the same number of non-overlapping windows by employing cyclic shifts from the top-left to the bottom-right regions, as illustrated in Fig. 2. This approach avoids extra computation caused by an increased number of windows. Finally, a hierarchical map is constructed by the patch merging layer; specifically, the model concatenates each group's feature maps of 2×2 neighboring patches from the previous stage and then puts the result in the next Swin Transformer blocks. The formal formulation for two consecutive transformer blocks is presented as follows:

$$\tilde{k}^l = \text{W-MSA}(\text{LN}(k^{l-1})) + k^{l-1}, \quad (28)$$

$$k^l = \text{MLP}(\text{LN}(\tilde{k}^l)) + \tilde{k}^l, \quad (29)$$

$$\tilde{k}^{l+1} = \text{SW-MSA}(\text{LN}(k^l)) + k^l, \quad (30)$$

$$k^{l+1} = \text{MLP}(\text{LN}(\tilde{k}^{l+1})) + \tilde{k}^{l+1}. \quad (31)$$

Suppose the input image has the size of $h \times w$ patches, and each window contains $M \times M$ patches; the computational complexity of a global MSA and the window-based one can be computed according to [34]:

$$\Omega(\text{MSA}) = 4hwC^2 + 2(hw)^2C, \quad (32)$$

$$\Omega(\text{W-MSA}) = 4hwC^2 + 2M^2hwC. \quad (33)$$

With the former scaling quadratically with the number of patches hw , and the latter scaling linearly when M is fixed. C is the dimension of the patch. The linear computation complexity of the Swin Transformer makes it suitable for transmitting high-resolution images captured from the satellite. For the transmitter, the output of the final block will be encoded by the channel encoder. At the receiver site, the semantic decoder will take the output of the channel decoder as input and reconstruct the image. Thus, instead of merging patches like in

the encoder, the architecture of the semantic decoder will be inversed of the encoder, such as patch splitting instead of patch merging. We consider the difference in computing capacity between the GU and BS by stacking more transformer blocks for the BS while keeping the minimum Swin transformer blocks for all GU devices.

Algorithm 1 Training Process for the Proposed SC System

- 1: **Initialize:** Satellite Encoder (E_α, C_β), Gateway Decoder ($C_{\phi_g}^{-1}, D_{\theta_g}$), GU u Decoder ($C_{\phi_u}^{-1}, D_{\theta_u}$), training epochs for each decoder T .
- 2: **Satellite & Gateway Training:**
- 3: **for** each epoch $t = 1, 2, \dots, T$ **do**
- 4: **for** each image batch **do**
- 5: Sampling noise N based on the SNR value δ ,
- 6: $X_I = C_\beta(E_\alpha(I), \delta)$,
- 7: $Y_{I,g} = H_{I,g}X_I + N$,
- 8: $\hat{I} = D_{\theta_g}(C_{\phi_g}^{-1}(Y_{I,g}))$,
- 9: $\mathcal{L} = \text{MSE}(I, \hat{I})$,
- 10: Update $E_\alpha, C_\beta, C_{\phi_g}^{-1}, D_{\theta_g}$ with $\nabla \mathcal{L}$.
- 11: **end for**
- 12: **end for**
- 13: **Training GU parameters:** Similar to the above training process, but only GU parameters are updated by the loss.
- 14: **Output:** Trained SC system.

2) **Channel Coder:** We implement a simple yet effective architecture for the channel coders to compress the extracted semantic features from the source encoder, which includes a number of fully connected layers. Additionally, while SNR information cannot reveal the exact noise, it does indicate the noise range, which is extremely useful for combating environmental noise [30], [45]. Therefore, we incorporate it into the channel encoder network by passing it through three fully connected layers before combining it with the encoded semantic features using an element-wise operation. The detailed design of the channel encoder is provided in Fig. 3. As shown in the Algorithm 1, we first train the satellite

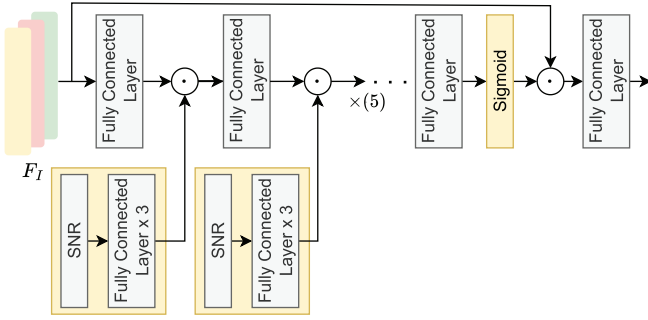


Fig. 3. The simple architecture of the channel encoder, which is built of fully connected layers and a skip connection.

and the gateway parameters, then train the GU parameters to adapt to the satellite network.

Algorithm 2 The Sequence to Train Denoising Module

- Step 1:** Train all the encoders and decoders of Satellite, Gateway, and GU with Algorithm 1.
- Step 2:** Encode, transmit and decode an image over a noisy environment two times with the same SNR value but different noise values to create different versions of noise images.
- Step 3:** Train the autoencoder-based denoising module with the multi-view images.

3) Autoencoder-Based Denoising Module: The autoencoder structure is famous for its ability to learn the latent representation [46], which motivates us to embrace architecture as our denoising model. The autoencoder model consists of the sub-module: an encoder and a decoder. In particular, the encoder projects the noisy image into low-dimension space to learn the latent representations. Then, the decoder part is leveraged to generate the noiseless image from the learned representations. The network parameters are trained using the MSE loss between the output of the decoder and the original image; through the training process, the encoder identifies and extracts the most important representations, and the decoder can reconstruct noiseless images by using the learned latent representations [47]. Additionally, the effects of the noise also become lighter due to information reduction during the encoding process. In addition, we acknowledged that the randomness of the noise from the environment will affect the transmitting signal in different ways, thus leading to different reconstructed images that even transmit the same signal under the same SNR. Leveraging this phenomenon, we propose a multi-view denoise technique that combines different noised images with the same SNR value to estimate the original one by minimizing the averaged mean square error among noised images toward the originally transmitted one. The process can be found in Algorithm 2, and it is worth noticing that the autoencoder is trained by the algorithm multiple times with different SNR values to adapt to varying wireless channel conditions.

The denoising module empowers the satellites to effectively suppress environmental noise in signals transmitted over

extended distances. A key benefit of our denoising model is its independence from end-to-end SC training. This enables its seamless deployment across various devices without the need for retraining. The proposed sequence is illustrated in Fig. 4.

B. Pathways and Subcarriers Selection to Meet QoS Demand

1) Sub-Problems: With the trained SC system, the proposed latency problem remains influenced by two key factors: the communication pathway and subcarrier allocation. The communication pathway selection, whether direct transmission or indirect through the gateway, has a significant role in facilitating the QoS for GUs. Meanwhile, the subcarrier allocation affects the transmission latency and is a discrete optimization problem. The solution for the combined problem highly depends on GU demands for image quality and physical environment conditions, which make the situation even more challenging to resolve. The pathways and subcarriers selection problem is described as follows:

$$\mathbf{P2:} \min_{\gamma, \chi} O(\gamma, \chi) \quad (34a)$$

$$\text{s.t. } \gamma_m^k \in \{0, 1\}, \quad \forall k \in \mathcal{K}, \forall m \in \mathcal{M} \quad (34b)$$

$$\sum_{m=1}^M \gamma_m^k \leq 1, \quad \forall k \in \mathcal{K} \quad (34c)$$

$$\sum_{k=1}^K \gamma_m^k \leq 1, \quad \forall m \in \mathcal{M} \quad (34d)$$

$$\chi_{u'}^j \in \{0, 1\}, \quad \forall j \in \mathcal{J}, \forall u' \in \mathcal{U} - u \quad (34e)$$

$$\sum_{u'=1}^{U-u} \chi_{u'}^j \leq 1, \quad \forall j \in \mathcal{J} \quad (34f)$$

$$\sum_{j=1}^J \chi_{u'}^j \leq 1, \quad \forall u' \in \mathcal{U} - u \quad (34g)$$

$$\mathbb{E}(\text{PSNR}_u(\mathbf{X})) \geq \Psi_u, \quad \forall u \in \mathcal{U} \quad (34h)$$

$$\mathbb{E}(\text{PSNR}_{u'}(\mathbf{X})) \geq \Psi_{u'}, \quad \forall u' \in \mathcal{U} - u' \quad (34i)$$

$$0 \leq \gamma^k t_{s,m} \leq T_m, \quad \forall m \in \mathcal{M}, k \in \mathcal{K}. \quad (34j)$$

Each GU has different requirements for image quality and experiences different noise levels. Some demand higher quality than others, while some are subjected to higher noise levels. By hopping the signal through the gateway for high-demand users, we can guarantee the QoS for those users, while the low-demand users can receive a signal from the direction pathway. We leverage the Discrete Whale Optimization Algorithm (DWOA) to determine the optimal pathways and sub-carrier frequency for GUs.

2) Discrete Whale Optimization Algorithm (DWOA) Path Selection & Subcarrier Allocation: We solve both problems with an effective algorithm. Specifically, we develop a two-stage DWOA to solve the pathways and subcarriers problem from the satellite to the GUs/gateway in the first stage. Then the second stage is to allocate the subcarrier frequency from the gateway to GUs. The problem formulation for satellite to gateways is given in (35):

$$\mathbf{P2.1:} \min_{\gamma} O(\gamma) \quad (35a)$$

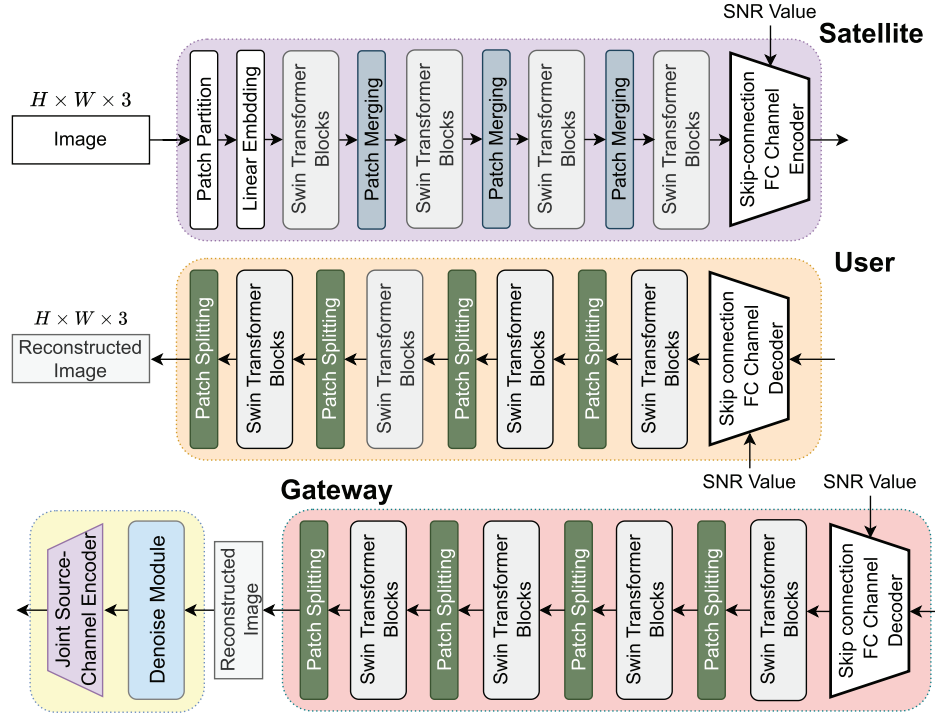


Fig. 4. The architecture of the joint source-channel encoder/decoder of the satellite, GU, and the gateway.

$$\text{s.t. } \gamma_m^k \in \{0, 1\}, \quad \forall k \in \mathcal{K}, \forall m \in \mathcal{M} \quad (35b)$$

$$\sum_{m=1}^M \gamma_m^k \leq 1, \quad \forall k \in \mathcal{K} \quad (35c)$$

$$\sum_{k=1}^K \gamma_m^k \leq 1, \quad \forall m \in \mathcal{M} \quad (35d)$$

$$\mathbb{E}(\text{PSNR}_u(\mathbf{X})) \geq \Psi_u, \quad \forall u \in \mathcal{U} \quad (35e)$$

$$0 \leq \gamma^k t_{s,m} \leq T_m, \quad \forall m \in \mathcal{M}, \quad (35f)$$

The DWOA is a powerful meta-heuristic technique designed to efficiently solve complex optimization problems involving integer decision variables, as encountered in our satellite-ground communication system. Specifically, the path selection between satellites and ground users (GUs) influences not only the transmission latency but also the quality of the reconstructed satellite images. As shown in constraints (35e), GUs have varying service quality demands, which necessitates distinct path selections for each GU. To address the challenge of integer-based decision variables in the optimization problems (35), we adopt the DWOA [48]. This algorithm adapts the traditional WOA to handle integer decision variables effectively. The DWOA initializes a population of candidate solutions (i.e., integer-based paths) within the feasible search space, iteratively refining these solutions through distinct exploration and exploitation phases which are defined in the subsequent sections.

a) Exploitation Phase: During the exploitation phase, DWOA mimics the bubble-net hunting strategy of humpback whales. The algorithm focuses on refining the current best solutions by adjusting them based on their proximity to other high-quality solutions. The update of the candidate solution

$\vec{\gamma}(\tau + 1)$ as:

$$\vec{\gamma}(\tau + 1) = \begin{cases} \vec{\gamma}^*(\tau) - \vec{A} \cdot \vec{D}_{Exp}, & p < 0.5, \\ \vec{D}' \cdot e^{bl} \cdot \cos(2\pi l) + \vec{\gamma}^*(\tau), & p \geq 0.5, \end{cases} \quad (36)$$

where $\vec{\gamma}^*(\tau)$ denotes the best solution at iteration τ , p is a random variable controlling the balance between two behaviors (shrinking encircling and spiral), $\vec{A}$ and $\vec{D}_{Exp}$ are coefficient vectors used to guide the exploitation process, $\vec{D}'$ denotes the distance from the current solution to the best solution obtained so far, b is a constant for defining the shape of the logarithmic spiral, and l represents a random number in $[-1, 1]$. The first condition mimics the whale's encircling behavior, while the second accounts for the whale's spiral movement toward prey. The vectors $\vec{A}$ and $\vec{D}_{Exp}$ are calculated as follows:

$$\vec{A} = 2\vec{a} \cdot \vec{r} - \vec{a}, \quad (37)$$

$$\vec{D}_{Exp} = |\vec{C} \cdot \vec{\gamma}^*(\tau) - \vec{\gamma}(\tau)|, \quad (38)$$

where $\vec{a}$ is gradually reduced in a linear manner from 2 to 0 throughout the iterations, $\vec{r}$ is a random vector in $[0, 1]$ as indicated in [49], and $\vec{C}$ is the coefficient vector, denoted as:

$$\vec{C} = 2\vec{r}. \quad (39)$$

b) Exploration Phase: In the exploration phase, DWOA promotes diversity in the population by encouraging candidate solutions to search the space globally. Each candidate solution moves towards a randomly selected solution in the population, allowing the algorithm to explore new areas of the search space. This can be mathematically described as:

$$\vec{\gamma}(\tau + 1) = \vec{\gamma}_{rand}(\tau) - \vec{A} \cdot \vec{D}_{Explr}. \quad (40)$$

where $\vec{\gamma}_{rand}(\tau)$ is the position of a randomly selected solution at iteration τ , and the coefficient vectors $\vec{A}$ and $\vec{D}_{Explr}$ are updated to control the step size and direction of exploration. While the $\vec{D}_{Explr}$ are as follows:

$$\vec{D}_{Explr} = |\vec{C} \cdot \vec{\gamma}_{rand}(\tau) - \vec{\gamma}(\tau)|. \quad (41)$$

c) *Fitness Evaluation & Constraint Handling*: At each iteration, the fitness of the candidate solutions is evaluated based on the objective function, which in our case is the minimizing transmission latency while ensuring the image quality for each user, as described by the PSNR constraints. To handle constraint violations, the algorithm incorporates a penalty function into the objective function. This ensures that infeasible solutions are penalized, guiding the search toward the feasible region of the solution space. The fitness function for the satellite-to-GU or gateway path selection and subcarrier allocation problem can be defined as follows:

$$f(\gamma) = O(\gamma) + \nu * F, \quad (42)$$

where ν defines the coefficient for violating the QoS and F denotes number of violations.

Algorithm 3 Discrete Whale Optimization Algorithm for SC-QoS

```

1: Input:  $\mathcal{N}$ : Search agents,  $\mathcal{K}$ , Pathway options & Subcarrier sets, SNR values,  $MaxIT$ .
2: Initialize whale population:  $\vec{\gamma}_k^n$  (Stage 1). Calculate initial latencies and set leading agent  $\vec{\gamma}_k^*$ ,  $\tau = 0$ .
3: while  $\tau \leq MaxIT$  do
4:   for  $n \leftarrow 1$  to  $\mathcal{N}$  do
5:     Update  $\vec{a}$ ,  $\vec{A}$ ,  $\vec{C}$ ,  $l$ ,  $p$  for new position calculation.
6:     if  $p < 0.5$  then
7:       if  $|A| < 1$  then
8:         Exploitation:  $\vec{\gamma}_k^n = \vec{\gamma}_k^* - \vec{A} \cdot \vec{D}_{Exp}$ 
9:       else
10:        Exploration:  $\vec{\gamma}_k^n = \vec{\gamma}_{rand} - \vec{A} \cdot \vec{D}_{Explr}$ 
11:      end if
12:    else
13:      Exploitation (spiral):  $\vec{\gamma}_k^n = \vec{D}' \cdot e^{bl} \cdot \cos(2\pi l) + \vec{\gamma}_k^*$ 
14:    end if
15:  end for
16:  Update latencies and best solution  $\vec{\gamma}_k^*$  if improved.
17:   $\tau \leftarrow \tau + 1$ 
18: end while
19: Output: Optimal pathways and subcarrier allocation:  $\vec{\gamma}_k^*$ .
```

d) *Algorithm Complexity*: The computational complexity of the DWOA can be expressed as [50]:

$$O(MaxIT * N * (2D + f(\gamma))) \quad (43)$$

where $MaxIT$ denotes the maximum number of iterations, N is the population size, D represents the dimensionality of the problem (number of GUs), and $f(\gamma)$ is the complexity of evaluating the fitness function. When the network scales in the number of GUs - D increases, the computation complexity of the algorithm linearly increases as shown in the equation (43), which demonstrates its computational efficiency and

adaptability for large networks. A comprehensive overview of the solution approach is presented in Algorithm 3.

Similarly, the problem formulation for gateways to bad SNR GUs is given in (44):

$$\text{P2.2: } \min_{\chi} O(\chi) \quad (44a)$$

$$\text{s.t. } \chi_{u'}^j \in \{0, 1\}, \quad \forall j \in \mathcal{J}, \quad \forall u' \in \mathcal{U} - u \quad (44b)$$

$$\sum_{u'=1}^{U-u} \gamma_{u'}^j \leq 1, \quad \forall j \in \mathcal{J} \quad (44c)$$

$$\sum_{j=1}^J \gamma_{u'}^j \leq 1, \quad \forall u' \in \mathcal{U} - u \quad (44d)$$

$$\mathbb{E}(\text{PSNR}_{u'}(\mathbf{X})) \geq \Psi_{u'}, \quad \forall u' \in \mathcal{U} - u \quad (44e)$$

Given the identical objective function and constraints of problems (44) and (35), the DWOA algorithm, previously employed for (35), is also applicable to (44). To avoid redundancy, the comprehensive solution algorithm is provided in the subsequent subsection.

Algorithm 4 2-Stage Pathway Selection Using DWOA for SC-QoS Problem

- 1: **Input:** Γ : Environment conditions, $\Psi_u, \Psi_{u'}$: QoS demands, T_m : access window.
 - 2: **Stage 1:** Use DWOA to select communication pathways and allocate subcarriers for direct satellite-to-GU/gateway communication, minimizing latency while meeting QoS.
 - 3: **Stage 2:** For indirect connections via gateways, DWOA optimizes subcarrier allocation from the gateway to the bad SNR GUs to minimize total latency and meet QoS demands.
 - 4: **Output:** Optimal pathways and subcarrier allocations: γ^* and χ^* for direct and indirect links, respectively.
-

3) *DWOA Implementation for SC-QoS, Path Selection & Subcarrier Allocation Optimization*: We apply the DWOA in a two-stage pathway selection algorithm for subcarrier allocation and path optimization in our satellite-ground communication system. The complete procedure is described in Algorithm 4. The optimal pathways and subcarriers γ^* and χ^* are determined for both direct and indirect communication links. By leveraging the discrete nature of DWOA, our approach efficiently tackles the integer constraints in path selection and resource allocation, yielding high-quality solutions for satellite-ground communication systems in 6G NTN.

C. Case Study

In this sub-section, we discuss deploying the proposed framework in real-world scenarios. With wide coverage capability and an appropriate altitude, LEO satellites are highly suitable for applications in earth surface observation, such as disaster warnings, victim rescues, and tracking moving objects. Taking the example of forest fire control, it is a very urgent task that requires high-quality images of affected regions and real-time data due to its fast spread rate. The deployment of UAVs will be limited by their energy and small coverage ability. Compared to UAVs, which consume significant energy

TABLE I
SIMULATION PARAMETERS

Notation	Definition	Value
$\mathcal{U}$	Number of GUs, i.e., $u \cup u'$	20
f_s^k	Satellite subcarrier frequency	[20 – 30] GHz [53]
B_s	Bandwidth of satellite subcarrier	500 MHz
G_s	Satellite antenna gain	33.13 dBi [53]
G_u	GU antenna gain	10.4 dBi [54]
G_g	Gateway antenna gain	59.0 dBi [55]
P_s	Satellite transmit power	10 W
d	Satellite altitude	786 km
T_k	Access window	60 sec
N_0	Satellite & GUs noise power	$\mathcal{N}(-44, 1)$ dB
B_g	Gateway subcarrier frequency	15 – 20 GHz
P_g	Gateway transmit power	1 W
N_1	Gateway & GUs noise power	$\mathcal{N}(-33, 2)$ dB
d_g	Gateway & GUs distances	10 km
SCR	Semantic compression ratio	1/16

for flight, LEO satellites can provide broader coverage and primarily dedicate their energy to communication and operational functions. The semantic communication system is deployed at the satellite can significantly reduce the communication latency by transmitting the important semantic feature to GUs. In addition, the gateway assistance can further improve the quality of the reconstruction image and low latency, which provides real-time information to firefighters, enabling them to extinguish fires effectively and rescue individuals in danger.

VI. SIMULATION RESULTS & DISCUSSION

A. Simulation Setup, Training & Evaluating Dataset

To validate the proposed gateway-assisted SC framework and assess the effectiveness of the two-stage DWOA, we conducted a series of simulations designed to highlight the advantages of our system. The simulation setting is carefully established by considering different simulation settings from various References as shown in Table I, with benchmark schemes introduced for comparative analysis. We experimented with both AWGN and Rayleigh fading channels across a wide range of noise power levels. Notably, the noise power between the satellite and the joint set of M devices was observed to be lower than that between the gateway and GUs, primarily due to the presence of obstacles in the latter communication link. However, the close distance between gateway and ground user and better antenna gain, the transmission through the gateway assistance is effective and guarantee the QoS for GUs. We utilize high-resolution images from the *DIV2K* training dataset [51], which consists of 800 images, to train the SC network. The performance of the trained network is then evaluated using both the *DIV2K* validation dataset (100 images) and the *FloodNet* dataset [52] to comprehensively demonstrate the effectiveness of the proposed approach. For the training process, a learning rate of $5e^{-4}$ is employed, and the Adam optimizer is utilized to optimize the deep learning models.

B. Benchmarks for Performance Comparison

To evaluate the performance improvements achieved through gateway-assisted SC and subcarrier frequency allocation in satellite networks, we consider the following benchmark schemes for comparison:

- *Direct Communication & Prioritize Algorithm*: In this approach, the satellite establishes direct connections with all GUs in the network, and the users with high-quality demand are prioritized (DI-PRI).
- *Direct Communication & Greedy Algorithm*: In this approach, the satellite establishes direct connections with all GUs in the network, and subcarrier frequency allocation is determined using the greedy algorithm (DI-GRE).
- *Direct Communication & DWOA*: Similar to the previous benchmark, the satellite directly communicates with all GUs; however, the subcarrier frequencies are allocated using the DWOA (DI-DWOA).
- *Hop Communication with Gateway Assisted & Prioritize Algorithm*: In this scheme, the satellite connects indirectly to GUs with low SNR through hop routing via a gateway. The high-demand users are prioritized to get assistance from the gateway (GA-PRI).
- *Hop Communication with Gateway Assisted & Greedy Algorithm*: In this scheme, the satellite connects indirectly to GUs with low SNR through hop routing via a gateway. The allocation of subcarrier frequencies is handled by the GRE (GA-GRE).

Due to the *cliff effect* of traditional communication, where the performance degrades significantly under an SNR threshold [28], [29]. In addition, the quality of the reconstruction image depends on the source encoder's compression rate, which operates independently of channel encoding, limiting performance gains under favorable channel conditions in traditional communication scenarios. It is impossible for it to accommodate the different quality demands of users, even with the gateway assistance, unless it adjusts the size of the transmitted signal. Therefore, we mainly focus on semantic communication, which leverages the joint-source channel coding design to meet the QoS requirements of ground users.

C. Results Analysis

1) *The Advantages of Gateway Assistance & Subcarrier Allocation*: In Fig. 5, we present a comparison of communication time between the direct connection scenario and the gateway-assisted approach. In the gateway-assisted configuration, the satellite establishes direct connections with half of the GUs, while the remaining half receives signals through a hop technique via the gateway. The results demonstrate that the gateway-assisted scenario significantly reduces the communication time for GUs compared to the direct connection approach. Notably, in the *DI-DWOA*, *DI-GRE*, and *DI-PRI* scenarios, communication time varies widely among GUs due to differences in the SNR between the satellite and each GU, which heavily influences transmission time. Given the same noise power N_0 , the gateway achieves a higher SNR than the GUs, benefiting from its enhanced receiving antenna gain, as shown in equation (8). Consequently, GUs in

TABLE II
THE PSNR REQUIREMENT OF USERS AND THE SECURED VALUES UNDER DIFFERENT SCENARIOS

User	1	2	3	4	5	6	7	8	9	10
<i>PSNR Requirement</i>	29.8864	29.8761	30.3789	30.1366	30.1479	29.8145	30.0042	29.8803	29.9676	30.2512
Achieved PSNR GA-DWOA	31.3657	32.7236	32.0022	30.9989	30.5937	32.0022	30.9989	31.7009	32.5127	30.9989
Achieved PSNR GA-GRE	34.2335	29.6511	31.3657	30.1458	30.5937	31.3657	30.5937	30.9989	32.2722	33.6661
Achieved PSNR GA-PRI	30.9989	29.6511	29.6511	30.9989	31.3657	29.1045	30.9989	29.1045	30.1458	30.5937
Achieved PSNR DI-DWOA	30.3161	30.3161	29.35	30.3161	30.3161	29.8556	30.3161	29.35	29.8556	30.3161
User	11	12	13	14	15	16	17	18	19	20
<i>PSNR Requirement</i>	29.9858	30.4977	29.7963	30.3045	29.8215	30.2775	29.9135	29.7610	30.2555	30.0924
Achieved PSNR GA-DWOA	32.5127	31.3657	31.7009	32.7236	31.7009	31.3657	32.2722	30.9989	30.5937	31.3657
Achieved PSNR GA-GRE	29.6511	33.8828	29.6511	32.0022	30.9989	30.5937	29.6511	30.5937	29.6511	34.3126
Achieved PSNR GA-PRI	30.1458	31.3657	29.6511	30.1458	29.1045	31.3657	29.6511	30.5937	30.5937	30.9989
Achieved PSNR DI-DWOA	30.3161	30.7371	29.8556	29.35	29.35	30.3161	29.35	30.3161	30.3161	30.3161

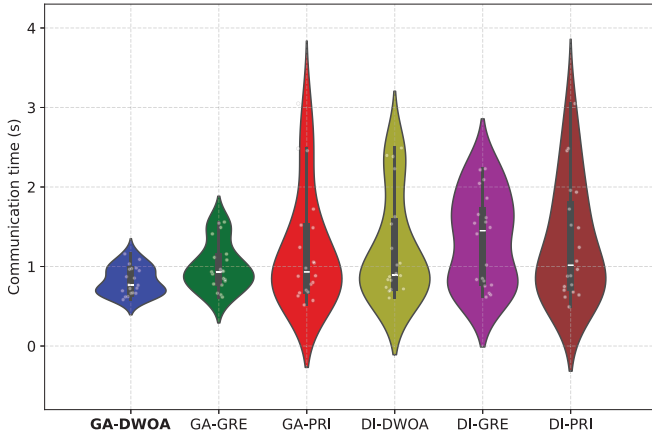


Fig. 5. Comparative analysis of communication times for GUs in direct and gateway-assisted scenarios employing DWOA, greedy algorithm, and prioritize algorithm.

the gateway-assisted scenario experience more consistent and reduced communication times compared to direct communication. Furthermore, the proposed two-stage *DWOA* algorithm outperforms the *GRE* and *PRI* algorithms in terms of transmission latency, achieving an average latency of 0.8186 seconds compared to 1.0008 and 1.0403 seconds of *GRE* and *PRI*, respectively.

2) *Importance of PSNR Threshold*: In Table II, we provide the QoS requirements for GUs and the corresponding Peak SNR (PSNR) performance across various scenarios. Each GU is assigned a unique QoS requirement for the received image, drawn from a normal distribution with a mean of 30 and a variance of 0.2. Among the benchmark approaches, only the proposed *GA-DWOA* successfully meets the QoS requirements for all GUs. In contrast, the *GA-GRE* approach prioritizes minimizing transmission time for each GU but fails to account for the overall system performance, resulting in penalties for QoS violations. In the meantime, the *GA-PRI* algorithm dedicates

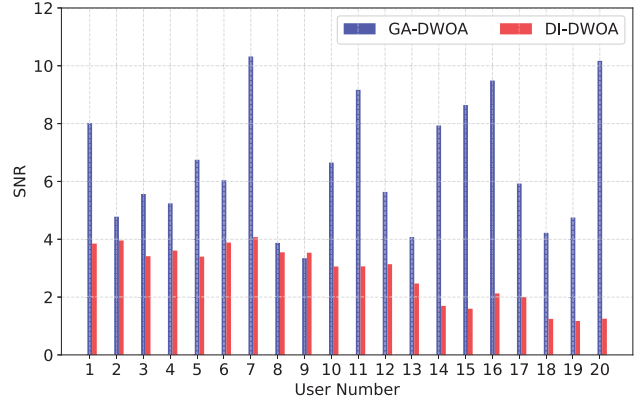


Fig. 6. Each GU channel quality for successful message transmission.

gateway assistance for high-demand users without considering channel conditions, which leads to the underperformance of other users, even those with lower quality demands, such as users 2, 13, and 15. Specifically, we impose a 0.5-second penalty to the communication latency for each QoS violation, representing the additional information needed to satisfy GU preferences. In scenarios where direct communication is employed between the satellite and GUs, even the *DWOA* algorithm cannot meet the QoS requirements for all GUs. The subcarrier frequency allocation determined by *DWOA* fails to satisfy the QoS for five GUs. These shortcomings highlight the inherent limitations of addressing the subcarrier frequency allocation problem without the benefit of gateway assistance. The results presented in the table demonstrate the effectiveness of the proposed gateway-assisted approach and underscore the critical role of subcarrier frequency allocation in maintaining image quality for GUs.

3) *Impact of Channel Conditions*: In Fig. 6, we demonstrate the channel condition impact (i.e., SNR) for each GU, with subcarrier frequencies optimized by *DWOA* in two

TABLE III

PSNR AND MS-SSIM PERFORMANCE OF GU, GATEWAY, GATEWAY WITH DENOISE MODULE UNDER AWGN AND RAYLEIGH FADING CHANNEL

Channel Type	AWGN Channel						Rayleigh Fading Channel					
Received Model	GU		Gateway		Gateway&Denoise		GU		Gateway		Gateway&Denoise	
SNR value	PSNR	MS-SSIM	PSNR	MS-SSIM	PSNR	MS-SSIM	PSNR	MS-SSIM	PSNR	MS-SSIM	PSNR	MS-SSIM
1 dB	28.7946	0.9271	28.8807	0.9270	29.1045	0.9309	27.4347	0.9061	27.5441	0.9077	27.7571	0.9124
3 dB	29.8556	0.9464	29.9619	0.9472	30.1458	0.9493	28.1708	0.9253	28.2585	0.9266	28.4689	0.9300
5 dB	30.7371	0.9588	30.8415	0.9597	30.9989	0.9610	28.7268	0.9382	28.7890	0.9387	29.0160	0.9415
7 dB	31.4642	0.9672	31.5562	0.9679	31.7009	0.9689	29.1212	0.9465	29.1658	0.9466	29.4114	0.9488
9 dB	32.0508	0.9731	32.1337	0.9736	32.2722	0.9745	29.3953	0.9520	29.4291	0.9518	29.7158	0.9545
11 dB	32.5138	0.9774	32.5930	0.9777	32.7236	0.9785	29.5796	0.9556	29.6113	0.9553	29.9221	0.9583
13 dB	32.8659	0.9805	32.9466	0.9807	33.0690	0.9813	29.7065	0.9580	29.7325	0.9577	30.0607	0.9609
15 dB	33.1127	0.9825	33.1990	0.9827	33.3242	0.9833	29.7919	0.9594	29.8110	0.9592	30.1534	0.9625
17 dB	33.2809	0.9838	33.3716	0.9841	33.5036	0.9847	29.8467	0.9604	29.8625	0.9605	30.2149	0.9635
19 dB	33.3895	0.9846	33.4859	0.9850	33.6214	0.9855	29.8817	0.9610	29.8957	0.9611	30.2549	0.9642

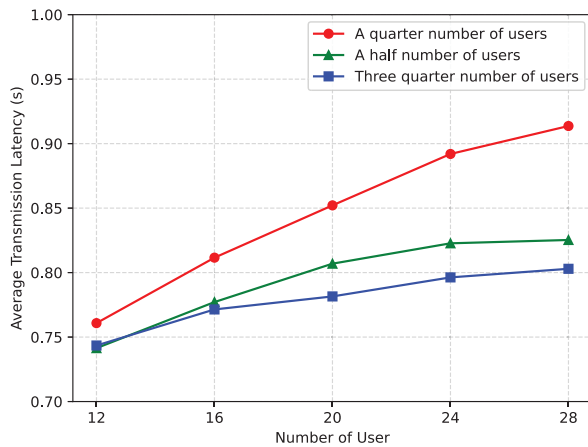


Fig. 7. Average transmission latency: Impact of GU Count and assisted gateway channel number.

scenarios: with and without gateway assistance (i.e., gateway-assisted and direct communication, respectively). Notably, GUs with low SNR values, such as GUs 14 to 20, are selected for hop and denoised communication via the gateway, where the higher antenna gains result in a significant SNR improvement. The indirect connection through the gateway not only boosts the SNR for GUs in high-noise environments (due to the antenna gain at the gateway) but also frees up better subcarrier frequencies for GUs with better initial conditions, allowing them to achieve even higher SNR values. For instance, GU 18, located in an area with noise power of $6.32e^{-8}$ watts, achieves an SNR of 1.246 with the optimized subcarrier allocation, whereas the same GU, when connected through the gateway using the same optimization algorithm, reaches an SNR of 4.222. A similar pattern is observed with GUs 19 and 20, while GU 9 experiences a slight reduction in SNR. This demonstrates the overall efficiency of the gateway-assisted approach in enhancing network performance by optimizing SNR for both low- and high-condition GUs.

4) *The Benefits of Gateway-Assisted Communication:* In our initial setup, we assumed that half of the total GUs, out of twenty, would receive assistance from the gateway. To assess the broader effectiveness of the proposed gateway assistance,

TABLE IV

PSNR AND MS-SSIM PERFORMANCE OF GU WHEN SUDDEN CHANGES OCCUR, WHICH LEADS TO THE MISMATCH BETWEEN ACTUAL SNR AND ESTIMATED SNR TO THE MODEL

Actual SNR=3dB	Estimated SNR=3dB	Estimated SNR=5dB
PSNR	29.8556	29.8411
MS-SSIM	0.9464	0.9457
Actual SNR=3dB	Estimated SNR=7dB	Estimated SNR=10dB
PSNR	29.6823	29.3462
MS-SSIM	0.9421	0.9343

we conducted experiments with varying numbers of GUs and different fractions receiving assistance: one-quarter, one-half, and three-quarters, as illustrated in Fig. 7. We also adjusted the number of subcarrier frequencies to correspond with the number of GUs, while maintaining consistent bandwidth. Our findings reveal that the average transmission latency for GUs increases as the number of GUs in the network grows, across all scenarios. However, the rate of latency increase varies significantly. In the scenario where only a quarter of GUs receive gateway assistance, the latency increases rapidly. In contrast, when half or three-quarters of GUs receive assistance, the latency increase is more gradual, with the three-quarter scenario achieving the lowest overall latency. Despite this, avoiding over-relying on gateway assistance is important, as the limited availability of subcarriers between the gateway and GUs may constrain performance. Therefore, the optimal level of gateway assistance should be carefully calibrated, based on the number of GUs experiencing low SNR, to balance transmission efficiency and resource availability effectively.

5) *Advantages of the Medium-Computing Decoder & Denoise Module at the Gateway:* In Table III, we present the average PSNR and Multi-Scale Structural Similarity Index Measure (MS-SSIM) metrics for GUs equipped with low-computing capabilities, the gateway with medium-computing power, and the gateway equipped with a denoising module, evaluated for both AWGN and Rayleigh fading channels. Across all models, performance improves as the SNR increases, with the most rapid gains observed between 1 and 13 dB, followed by more gradual improvements at higher SNR levels. At any given SNR value, the low-computing

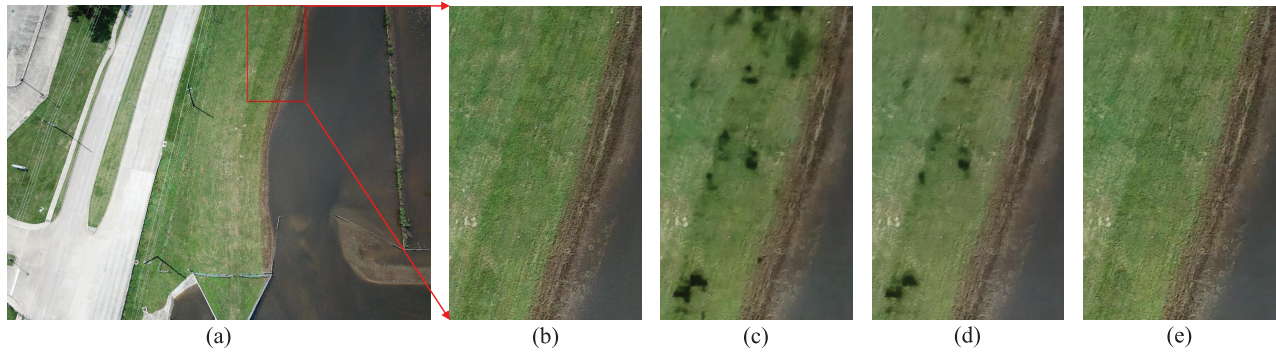


Fig. 8. (a) The original image, (b) The zoom-in region of original image, (c) The zoom-in region of direct transmission from satellite to the ground user with SNR = 0.359 dB, PSNR = 27.599, (d) The zoom-in region of reconstruction at the gateway with denoise module at SNR = 0.359 dB, PSNR = 28.342, (e) The zoom-in region of reconstruction image with gateway assistance and denoise module SNR = 2 dB, PSNR = 28.849.

TABLE V
SUMMARY OF KEY PERFORMANCE METRICS ACROSS BASELINE FRAMEWORKS

Communication Pathway	Direct Only			Gateway-assisted		
Algorithms	PRI	GRE	DWOA	PRI	GRE	DWOA
Computation Cost	Satellite and GUs only			Extra computation at Gateway		
Communication Latency	Average	High	Average	Average	Average	Low
Feasible PSNR Requirements	Partial	A Few	Partial	Partial	Partial	All

GUs consistently exhibit lower performance than the medium-computing gateway, particularly at low SNR levels, with a performance gain of approximately 0.1 in PSNR for both channels. However, the gateway equipped with the denoising module demonstrates significant performance gains, especially at lower SNR values. For instance, at 1 dB, the gateway's PSNR improves from 28.7946 to 29.1045 in the AWGN channel and from 27.4347 to 27.7571 in the Rayleigh channel. While the MS-SSIM gains of 0.0038 and 0.0063 may appear modest, they are substantial when considering that the maximum MS-SSIM value is 1. These results highlight the considerable advantage of employing a denoising module at the gateway, particularly in challenging communication environments.

6) *Analysis of Image Retrieval & Denoising:* In Fig. 8, we validated the effectiveness of the proposal with the reconstruction of the image at GU through both direct connection and gateway hop, as well as the image reconstructed at the gateway. We considered a challenging scenario to show the noise effects on the visualization aspect of the reconstruction image. Specifically, the noise power is equal to -40 dB, and the frequency of the subcarrier allocated to the user is 29.75 GHz, which results in the SNR value at 0.359 dB, and this noise creates several irregularly shaped black spots scattered on the reconstructed image (c) at the user. To highlight the capabilities of the denoising module at the gateway, we provided the resulting image (d) when it experienced the same SNR value. As we can see, there is a significant improvement between the subfigures (c) and (d), in which the noise regions become smaller and less severe. When the signal is hopped through the gateway, the SNR value is improved and reaches 2 dB due to the high antenna gain of the gateway, while the SNR between the gateway and the user is 5 dB. The reconstruction image under this scenario is illustrated in the final subfigure (e) in Fig. 8; the image has a much more visually pleasing

appearance with no visible dark spots compared with others. In general, by combining the advantages of the gateway in high-antenna gain (better SNR) and high-computing capabilities (image denoise module), we achieve higher PSNR metrics and better visual images, which can proactively facilitate the QoS of GUs.

7) *The Robustness of the Proposed Semantic Communication:* In a dynamic environment, the channel condition will inevitably encounter sudden change, resulting in mismatched values between the actual SNR of the environment and the informed SNR. Therefore, the semantic communication system needs to be robust to sudden changes in noise and interference. In Table IV, we illustrate the performance of GUs when there is a mismatch in information between the actual channel condition and the informed value under various cases. As we can see, the semantic communication performance of GU cannot achieve the optimal value, but it has relatively good results. The results of this table show that the proposed semantic communication model can be operated in a dynamic environment that encounters changes in the channel condition. This robustness ability is achieved by exposure of the model to various noise levels during the training process.

Table V compares the performance of the proposed frameworks against other benchmarks across various aspects. The gateway assistance will consume more computation cost compared to direct transmission. However, it lowers the system's communication latency and addresses the PSNR requirement of GUs, which outweighs the computation cost. In addition, the gateway has unlimited energy resources and high computing capacities compared to the GUs, rendering the additional computation cost negligible.

Lastly, we conduct experiments to train semantic communication models to output different semantic compression rates between the satellite and GUs in Fig. 9. As the length of the transmitted signal increases, the system achieves higher

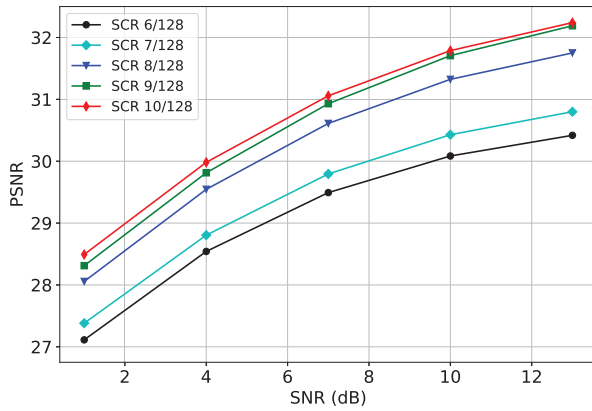


Fig. 9. The performance of semantic communication with different semantic compression rates under various channel conditions.

PSNR values, indicating higher similarity to the original image. When receiving a longer signal, the GU obtains more information about the original data, enabling accurate signal decoding and reducing the impact of environmental noise.

VII. CONCLUSION

In this paper, we address the challenges of SC in satellite networks, particularly in scenarios where multiple GUs experience degraded signal quality due to satellite altitude. To enhance communication efficiency and reliability, we proposed a novel gateway-assisted framework that leverages gateway capacity to mitigate environmental noise and improve signal-to-noise ratio (SNR) for GUs. The approach involved optimizing communication overhead while ensuring data reconstruction quality, using a two-stage DWOA. Our simulation results demonstrate that the gateway-assisted strategy significantly improves image quality in high-noise environments, even with transmission distortion. Additionally, DWOA outperformed traditional greedy algorithms by reducing transmission latency while maintaining QoS for GUs. Our findings represent a key advancement in satellite-based SC networks, offering more reliable and efficient data transmission, particularly in disaster relief and time-sensitive applications. The proposed system also effectively reduces bandwidth requirements, adapts to complex satellite scenarios, and protects semantic information with acceptable transmission delays. In future works, we can extend the idea in two directions. Firstly, we can consider the collaboration across multiple satellite constellations, where the data can be transmitted to multiple satellites by leveraging laser communication before being relayed to the ground gateway near the targeted GU and then to the targeted GUs. Secondly, the transmission of multi-modality data in semantic communication can be considered by adopting new state-of-the-art deep learning frameworks.

REFERENCES

- [1] S. Mahboob and L. Liu, "Revolutionizing future connectivity: A contemporary survey on AI-empowered satellite-based non-terrestrial networks in 6G," *IEEE Commun. Surveys Tuts.*, vol. 26, no. 2, pp. 1279–1321, 2nd Quart., 2024.
- [2] B. Di, L. Song, Y. Li, and H. V. Poor, "Ultra-dense LEO: Integration of satellite access networks into 5G and beyond," *IEEE Wireless Commun.*, vol. 26, no. 2, pp. 62–69, Apr. 2019.
- [3] O. Kotheli et al., "Satellite communications in the new space era: A survey and future challenges," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 1, pp. 70–109, 1st Quart., 2021.
- [4] K. Messaoudi, O. S. Oubbati, A. Rachedi, and T. Bendouma, "UAV-UGV-based system for AoI minimization in IoT networks," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Rome, Italy, May 2023, pp. 4743–4748.
- [5] O. S. Oubbati, H. Badis, A. Rachedi, A. Lakas, and P. Lorenz, "Multi-UAV assisted network coverage optimization for rescue operations using reinforcement learning," in *Proc. IEEE Consum. Commun. Netw. Conf. (CCNC)*, Jul. 2023, pp. 1003–1008.
- [6] J. Heo, S. Sung, H. Lee, I. Hwang, and D. Hong, "MIMO satellite communication systems: A survey from the PHY layer perspective," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 3, pp. 1543–1570, 3rd Quart., 2023.
- [7] S. S. Hassan, Y. M. Park, Y. K. Tun, W. Saad, Z. Han, and C. S. Hong, "SpaceRIS: LEO satellite coverage maximization in 6G sub-THz networks by MAPPO DRL and whale optimization," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 5, pp. 1262–1278, May 2024.
- [8] M. Hosseini, J. P. Choi, S.-H. Chang, and J. Lee, "Review of 5G NTN standards development and technical challenges for satellite integration with the 5G network," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 36, no. 8, pp. 22–31, Aug. 2021.
- [9] F. Fourati and M.-S. Alouini, "Artificial intelligence for satellite communication: A review," *Intell. Conver. Netw.*, vol. 2, no. 3, pp. 213–243, Sep. 2021.
- [10] W. Yang et al., "Semantic communications for future internet: Fundamentals, applications, and challenges," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 1, pp. 213–250, 1st Quart., 2023.
- [11] G. Shi, Y. Xiao, Y. Li, and X. Xie, "From semantic communication to semantic-aware networking: Model, architecture, and open problems," *IEEE Commun. Mag.*, vol. 59, no. 8, pp. 44–50, Aug. 2021.
- [12] H. Zhang, R. Liu, A. Kaushik, and X. Gao, "Satellite edge computing with collaborative computation offloading: An intelligent deep deterministic policy gradient approach," *IEEE Internet Things J.*, vol. 10, no. 10, pp. 9092–9107, May 2023.
- [13] B. A. Homssi et al., "Artificial intelligence techniques for next-generation massive satellite networks," *IEEE Commun. Mag.*, vol. 62, no. 4, pp. 66–72, Apr. 2024.
- [14] C. Chaccour, W. Saad, M. Debbah, Z. Han, and H. V. Poor, "Less data, more knowledge: Building next-generation semantic communication networks," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 1, pp. 37–76, 1st Quart., 2025.
- [15] H. Peng, Z. Zhang, Y. Liu, Z. Su, T. H. Luan, and N. Cheng, "Semantic communication in non-terrestrial networks: A future-ready paradigm," *IEEE Netw.*, vol. 38, no. 4, pp. 119–127, Jul. 2024.
- [16] J. Guo, D. Rincón, S. Sallent, L. Yang, X. Chen, and X. Chen, "Gateway placement optimization in LEO satellite networks based on traffic estimation," *IEEE Trans. Veh. Technol.*, vol. 70, no. 4, pp. 3860–3876, Apr. 2021.
- [17] M. Moldoveanu and A. Zaidi, "In-network learning: Distributed training and inference in networks," *Entropy*, vol. 25, no. 6, p. 920, Jun. 2023.
- [18] S. Yang, H. Pan, T.-T. Chan, and Z. Wang, "Semantic communication-empowered physical-layer network coding," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Glasgow, U.K., Mar. 2023, pp. 1–6.
- [19] C. Bian, Y. Shao, H. Wu, and D. Gunduz, "Deep joint source-channel coding over cooperative relay networks," 2022, *arXiv:2211.06705*.
- [20] X. Luo, B. Yin, Z. Chen, B. Xia, and J. Wang, "Autoencoder-based semantic communication systems with relay channels," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, Seoul, South Korea, May 2022, pp. 711–716.
- [21] D. Jiang et al., "QoE-aware efficient content distribution scheme for satellite-terrestrial networks," *IEEE Trans. Mobile Comput.*, vol. 22, no. 1, pp. 443–458, Jan. 2023.
- [22] A. I. Ameur, O. S. Oubbati, A. Lakas, A. Rachedi, and M. B. Yagoubi, "Efficient vehicular data sharing using aerial P2P backbone," *IEEE Trans. Intell. Vehicles*, early access, Jun. 13, 2024, doi: [10.1109/ITV.2024.3414140](https://doi.org/10.1109/ITV.2024.3414140).
- [23] S. Yu, X. Gong, Q. Shi, X. Wang, and X. Chen, "EC-SAGINs: Edge-computing-enhanced space-air-ground-integrated networks for Internet of Vehicles," *IEEE Internet Things J.*, vol. 9, no. 8, pp. 5742–5754, Apr. 2022.

- [24] S. S. Hassan, Y. M. Park, Y. K. Tun, W. Saad, Z. Han, and C. S. Hong, "Satellite-based ITS data offloading & computation in 6G networks: A cooperative multi-agent proximal policy optimization DRL with attention approach," *IEEE Trans. Mobile Comput.*, vol. 23, no. 5, pp. 4956–4974, May 2024.
- [25] C. Jiang and X. Zhu, "Reinforcement learning based capacity management in multi-layer satellite networks," *IEEE Trans. Wireless Commun.*, vol. 19, no. 7, pp. 4685–4699, Jul. 2020.
- [26] X. Luo, H. Chen, and Q. Guo, "Semantic communications: Overview, open issues, and future research directions," *IEEE Wireless Commun.*, vol. 29, no. 1, pp. 210–219, Feb. 2022.
- [27] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, May 2021.
- [28] E. Boursoulatz, D. B. Kurka, and D. Gündüz, "Deep joint source-channel coding for wireless image transmission," *IEEE Trans. Cogn. Commun. Netw.*, vol. 5, no. 3, pp. 567–579, Sep. 2019.
- [29] D. B. Kurka and D. Gündüz, "DeepJSCC-f: Deep joint source-channel coding of images with feedback," *IEEE J. Sel. Areas Commun.*, vol. 1, no. 1, pp. 178–193, May 2020.
- [30] J. Xu, B. Ai, W. Chen, A. Yang, P. Sun, and M. Rodrigues, "Wireless image transmission using deep source channel coding with attention modules," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 4, pp. 2315–2328, May 2022.
- [31] D. Huang, F. Gao, X. Tao, Q. Du, and J. Lu, "Toward semantic communications: Deep learning-based image semantic coding," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 1, pp. 55–71, Jan. 2023.
- [32] W. Zhang, H. Zhang, H. Ma, H. Shao, N. Wang, and V. C. M. Leung, "Predictive and adaptive deep coding for wireless image transmission in semantic communication," *IEEE Trans. Wireless Commun.*, vol. 22, no. 8, pp. 5486–5501, Aug. 2023.
- [33] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Long Beach, CA, USA, Dec. 2017, pp. 6000–6010.
- [34] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Montreal, QC, Canada, Oct. 2021, pp. 9992–10002.
- [35] P. Yi, Y. Cao, J. Xu, and Y.-C. Liang, "Semantic communication for remote spectrum sensing in non-terrestrial networks," in *Proc. IEEE 32nd Int. Workshop Mach. Learn. Signal Process. (MLSP)*, Xi'an, China, Aug. 2022, pp. 1–6.
- [36] G. Zheng, Q. Ni, K. Navaie, and H. Pervaiz, "Semantic communication in satellite-borne edge cloud network for computation offloading," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 5, pp. 1145–1158, May 2024.
- [37] Z. Zhao et al., "Energy-efficient probabilistic semantic communication over space-air-ground integrated networks," *IEEE Trans. Wireless Commun.*, early access, May 19, 2025, doi: [10.1109/TWC.2025.3569102](https://doi.org/10.1109/TWC.2025.3569102).
- [38] P. Jiang, C.-K. Wen, X. Li, S. Jin, and G. Y. Li, "Semantic satellite communications based on generative foundation model," *IEEE J. Sel. Areas Commun.*, early access, Apr. 9, 2025, doi: [10.1109/JSAC.2025.3559113](https://doi.org/10.1109/JSAC.2025.3559113).
- [39] S. S. Hassan, L. X. Nguyen, Y. K. Tun, Z. Han, and C. S. Hong, "Semantic enabled 6G LEO satellite communication for Earth observation: A resource-constrained network optimization," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Cape Town, South Africa, Dec. 2024, pp. 4472–4478.
- [40] D. Deng et al., "Semantic communication empowered NTN for IoT: Benefits and challenges," *IEEE Netw.*, vol. 38, no. 4, pp. 32–39, Jul. 2024.
- [41] W. Lu et al., "Reinforcement learning driven time-sensitive moving target tracking of intelligent agile satellite," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 6, pp. 9085–9101, Dec. 2024.
- [42] H. Xie, Z. Qin, X. Tao, and K. B. Letaief, "Task-oriented multi-user semantic communications," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 9, pp. 2584–2597, Sep. 2022.
- [43] H. Xie, Z. Qin, and G. Y. Li, "Semantic communication with memory," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 8, pp. 2658–2669, Aug. 2023.
- [44] Q. Zhou, R. Li, Z. Zhao, C. Peng, and H. Zhang, "Semantic communication with adaptive universal transformer," *IEEE Wireless Commun. Lett.*, vol. 11, no. 3, pp. 453–457, Mar. 2022.
- [45] K. Yang, S. Wang, J. Dai, K. Tan, K. Niu, and P. Zhang, "WITT: A wireless image transmission transformer for semantic communications," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Rhodes Island, Greece, Jun. 2023, pp. 1–5.
- [46] A. Creswell and A. A. Bharath, "Denoising adversarial autoencoders," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 4, pp. 968–984, Apr. 2019.
- [47] H. Chen, C. Qu, Y. Zhang, C. Chen, and J. Jiao, "Multi-view self-supervised disentanglement for general image denoising," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Paris, France, Oct. 2023, pp. 12247–12257.
- [48] Y. Li, Y. He, and X. Liu, "A novel discrete whale optimization algorithm for solving knapsack problems," *Appl. Intell.*, vol. 50, pp. 3350–3366, Jun. 2020.
- [49] S. Mirjalili and A. Lewis, "The whale optimization algorithm," *Adv. Eng. Softw.*, vol. 95, pp. 51–67, May 2016.
- [50] Q.-V. Pham, S. Mirjalili, N. Kumar, M. Alazab, and W.-J. Hwang, "Whale optimization algorithm with applications to resource allocation in wireless networks," *IEEE Trans. Veh. Technol.*, vol. 69, no. 4, pp. 4285–4297, Apr. 2020.
- [51] E. Agustsson and R. Timofte, "NTIRE 2017 challenge on single image super-resolution: Dataset and study," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 1122–1131.
- [52] M. Rahnmooonfar, T. Chowdhury, A. Sarkar, D. Varshney, M. Yari, and R. R. Murphy, "FloodNet: A high resolution aerial imagery dataset for post flood scene understanding," *IEEE Access*, vol. 9, pp. 89644–89654, 2021.
- [53] H.-H. Choi, G. Park, K. Heo, and K. Lee, "Joint optimization of beam placement and transmit power for multibeam LEO satellite communication systems," *IEEE Internet Things J.*, vol. 11, no. 8, pp. 14804–14813, Apr. 2024.
- [54] Y. Wang and F. Xu, "Shared-aperture 4G LTE and 5G mm-Wave antenna in mobile phones with enhanced mm-wave radiation in the display direction," *IEEE Trans. Antennas Propag.*, vol. 71, no. 6, pp. 4772–4782, Jun. 2023.
- [55] S. Haussmann et al., "Measurement and analysis of FDM for E-band satellite communication," in *Proc. 53rd Eur. Microw. Conf. (EuMC)*, Berlin, Germany, Sep. 2023, pp. 356–359.